An Exponential Succinctness Gap between Three-Variable Logic and the Calculus of Relations

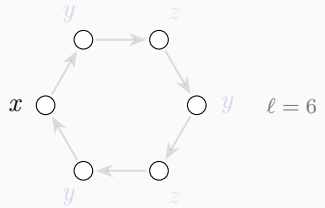
Yuya Uezato Yoshiki Nakamura

CyberAgent, Inc. Chiba University

STOC 2027 submission (author video for the PC)

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$\ell = 6$

$$\exists y\,(E(x,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge E(y,x))))))$$

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

☛ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]

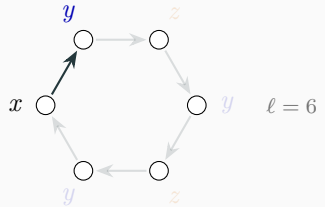
スライド概要：FO^k は変数を使い回せる。モデル検査は $O(|\varphi| n^k)$。

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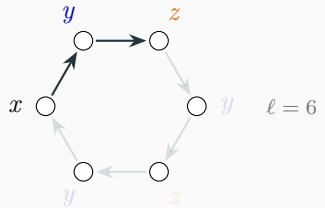
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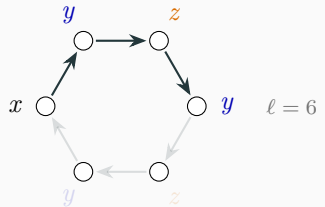
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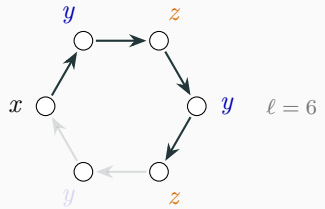
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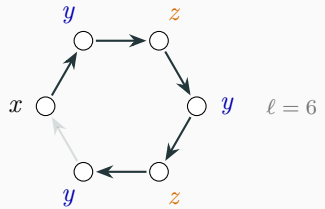
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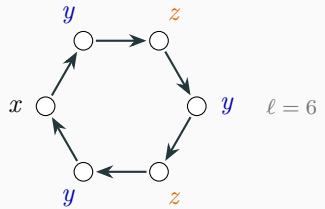
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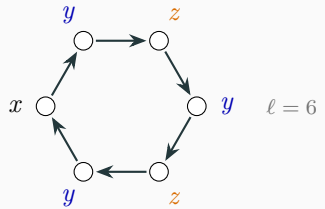
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FO^k model checking: $O(n^k)$ is 'optimal' for $k \geq 4$; the exception is FO^3

We consider binary relation structures with n elements:

syntax	model checking	
FO^k, $k \geq 4$	no $O(n^{k-\varepsilon})$ under HyperClique Hypothesis	[BS21, ICALP] [NU27, PODS]
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CoR [Tar41, JSL] Tarski's calculus of relations	$O(\lvert\varphi\rvert \cdot n^\omega)$	

Boolean matrix algorithm of FO^3 model checking via $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ [Tar41, JSL; TG87]:

$$\exists z.\ P(x,z) \wedge Q(z,y) \quad \rightsquigarrow \quad (P;Q)(x,y)$$

composition = Boolean matrix multiplication

To obtain the form of $\exists z.\ \cdots \wedge \cdots$, we use DNF translation.
$\rightsquigarrow 2^{O(\lvert\varphi\rvert)}$ compositions are used [Nak22].

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メモ：

- 合成 = ブール行列積
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Are $2^{O(|\varphi|)}$ compositions unavoidable?

DNF translation is used, for example:

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$$\rightsquigarrow_{\text{(DNF)}}\ \exists z.\ \bigvee_{A \subseteq [k]} \Big(\bigwedge_{i \in A} P_i(x,z) \wedge \bigwedge_{i \notin A} Q_i(z,y) \Big)$$

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(In this translation, we use 2^k compositions, since the number of subsets $A \subseteq [k]$ is 2^k.)

Problem (Open [FGL+15, Inf. Sci.; Nak22, JLAMP; NU27, PODS])

Can we prove that there is no polynomial-size translation from FO^3 to CoR?

Partial answer [Nak22, JLAMP]: yes for **existential** CoR [Nak23, LICS] ($\equiv$ existential FO^3), via preservation under extensions, which fails for full CoR.

スライド概要：この指数個の合成は必要か？ 未解決だった（$\Sigma_1\mathrm{CoR}$ のみ既知）。

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Outline

1. Our results

2. Technical overview

3. Conclusion

Our results

The answer is **YES!**

Thm. (This paper)

There is no polynomial-size translation from FO^3 *to* CoR.
More precisely, there are FO^3 *formulas* $\varphi_1, \varphi_2, \ldots$ *with* $|\varphi_k| = O(k)$ *such that every* CoR *term equivalent to* φ_k *needs* $\geq 2^{k-1}$ *compositions.*

Also in this paper: exponential lower bounds for

- $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$, binary — before:
  exponential gap FO^k vs. FO^{k-1} ($k \geq 4$), higher arity [VS24, LICS]
  $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$ is polynomial over linear orders [GS05, LMCS]
- ARA(3) $\rightsquigarrow$ matrix circuits — asked by [BGV19, arXiv; BGV22, AMAI]
- $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ with TC

👉 Our proof technique can apply to all the above!

スライド概要：答えは YES（主定理）。同じ手法で FO^2、行列言語、閉包付き CoR にも指数下界。

メモ：

- FO^2：$O(k)$ サイズの FO^3 文、同値な FO^2 文は量化子 $\geq 2^k$
- 行列：集約ゲート $\geq 2^k/(3\beta)$（β：半環の join breadth）
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Our results

The answer is **YES!**

Thm. (This paper)

There is no polynomial-size translation from FO^3 *to* CoR.
More precisely, there are FO^3 *formulas* $\varphi_1, \varphi_2, \ldots$ *with* $|\varphi_k| = O(k)$ *such that every* CoR *term equivalent to* φ_k *needs* $\geq 2^{k-1}$ *compositions.*

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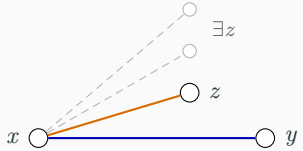
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The hard formula G_k

$$G_k(x,y) \coloneqq \exists z \bigwedge_{i=1}^{k} \left[(P_i(x,y) \wedge Q_i(x,z)) \vee (Q_i(x,y) \wedge P_i(x,z)) \right] \qquad (|G_k| = O(k))$$



we can pick z to satisfy: for every $i \in [k]$,
the x-y and x-z edges
have labels P_i and Q_i, in either order.

CoR: 2^k compositions

$$G_k \equiv \bigcup_{A \subseteq [k]} \left[\left(\bigcap_{i \in A} P_i \cap \bigcap_{i \notin A} Q_i \right) \cap \left(\left(\bigcap_{i \in A} Q_i \cap \bigcap_{i \notin A} P_i \right) ; \mathbf{1} \right) \right]$$

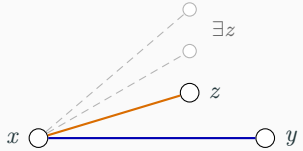
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Outline

1. Our results

2. Technical overview

3. Conclusion

Proof outline (at high-level)

Goal Lemma Every CoR term equivalent to G_k needs $2^{\Omega(k)}$ compositions.

We construct structures $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ such that

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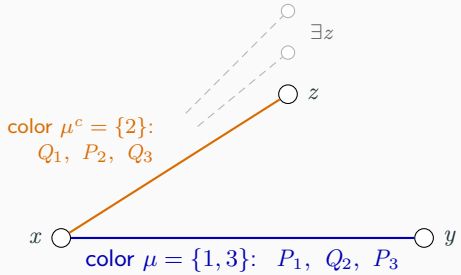
メモ：

- 表の T/F は、各 $\mathcal{M}_i$ に対応する点対での値（論文の test pair）
- 以降：$\mathcal{M}_0$ の作り方、$\mathcal{M}_i$ = hub での塗り替え $\mathcal{M}_\lambda$、合成が新たに見分けるのは高々 2 個である理由

Structures as colored complete graphs

View structures as **complete graphs**, colored with 2^k colors $\mu \subseteq [k]$: $P_i \iff i \in \mu$.

Q_i is used as the complement of P_i (i.e., $i \notin \mu$): exactly one of P_i, Q_i holds on each edge.



for every $i \in [3]$,
the x-y and x-z edges
have labels P_i and Q_i, in either order
$\Rightarrow$ $G_3(x, y)$ $(k = 3)$

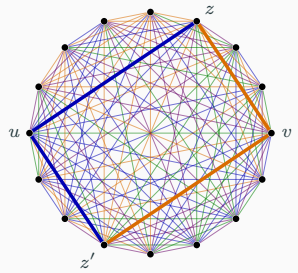
☛ $G_k(x, y)$ holds iff some x-z edge has the color μ^c, where μ is the color of the x-y edge.

スライド概要：構造を色付き完全グラフとして見る。G_k は「補の色の x-z 辺がある」と読める。

メモ：

- 各辺で P_i と Q_i のちょうど一方が成り立つ
- 例：$k = 3$、$\{1, 3\}$ と $\{2\}$ は互いに補

The randomly colored structure $\mathcal{M}_0$



$\mathcal{M}_0$: randomly colored, $O(k4^k)$ vertices
[JMT95, Algebra Univers.]

Two-witness property For all $u \neq v$ and all colors $\boldsymbol{\mu}, \boldsymbol{\nu}$, there are at least two **witnesses** z such that the u-z and z-v edges have colors $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$, respectively.

Every CoR term t is color-invariant on $\mathcal{M}_0$: $\mathcal{M}_0 \models t(u,v) \iff \mathcal{M}_0 \models t(u',v')$ whenever the u-v and u'-v' edges have the same color.

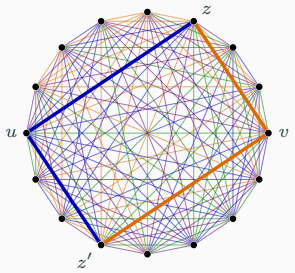
Proof by induction on t: the only nontrivial case, $t = t_1 \,;\, t_2$, follows from the (two-)witness property.

スライド概要：$\mathcal{M}_0$ ではどの色の道のパターンにも中間の頂点がある。だから項は「色 $\mapsto$ T/F」の表になる。

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- 「2 つ以上」が効くのは $\mathcal{M}_\lambda$ 側（hub を避ける中間の頂点が残る）

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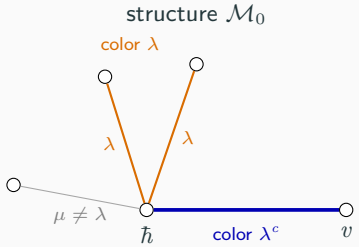
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$\mathcal{M}_\lambda$: recoloring at the hub $\hbar$



$G_k(\hbar, v) = 1$

$G_k(\hbar, v)$ needs a $\hbar$-z edge of color λ
recoloring λ changes **exactly those edges at** $\hbar$, and nothing else

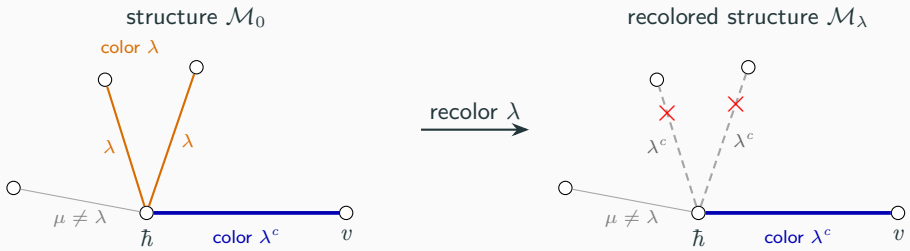
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away from $\hbar$, terms cannot distinguish $\mathcal{M}_\lambda$ from $\mathcal{M}_0$ (of the ≥ 2 middle vertices, one avoids $\hbar$)

スライド概要：塗り替えで G_k が 1 から 0 に変わる。

メモ：

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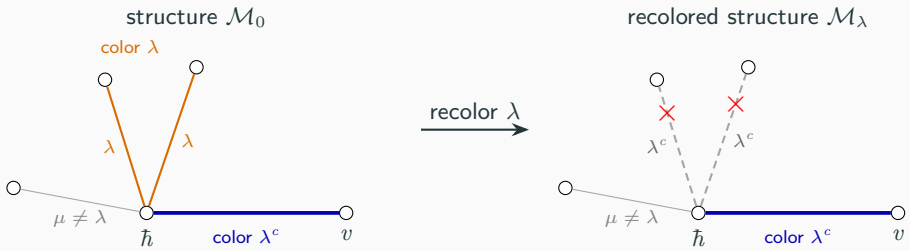
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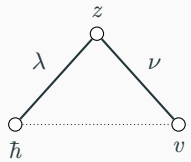
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Why a composition newly distinguishes at most two $\mathcal{M}_\lambda$

(a) $\mathcal{M}_0$, at $(\hbar, v)$



middle vertex z:
$(X\,;\,Y)(\hbar, v) = 1$

a composition $X\,;\,Y$, where X holds on the edges of color λ and Y on those of color $\nu \neq \lambda$

only pairs at the hub can change, and only if X or Y singles out the color λ

$\Rightarrow$ each composition newly distinguishes at most two $\mathcal{M}_\lambda$ from $\mathcal{M}_0$ (hence the Key Lemma)

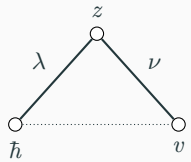
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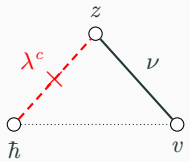
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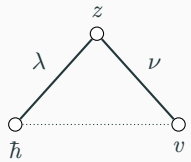
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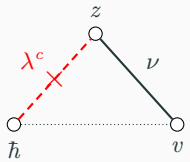
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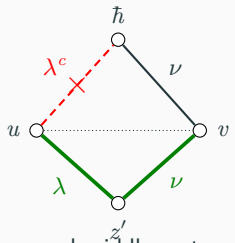
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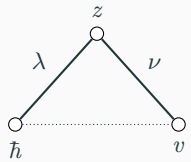
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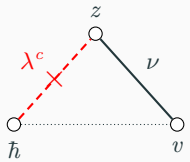
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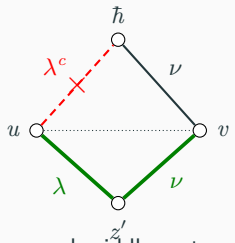
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Outline

1. Our results

2. Technical overview

3. Conclusion

Summary

Result Tarski's translation from FO^3 to CoR is optimal in the worst case:

- some FO^3 formulas of size $O(k)$ need $2^{\Omega(k)}$ compositions in every equivalent CoR term, even with complement, converse, and shared subterms;
- the same technique: FO^3 vs. FO^2, matrix query languages, CoR with closure.

Technique 2^k recolorings of one structure $\mathcal{M}_0$; each composition newly distinguishes ≤ 2 of them.

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- a superpolynomial gap over a **fixed** finite signature (e.g., one binary relation)?
- the domain size from which the blowup begins?
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Thank you for your attention!

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メモ：

- 未解決：固定シグネチャ、開始の領域サイズ、正確な合成数

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